

Selling Training Data

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1 Introduction

Selling Training Data to Agents with Private Dataset

Data is increasingly traded as an input into prediction and decision-making.

AI training · credit and insurance · healthcare and genomics · digital advertising

A buyer typically enters the data market with an existing dataset P'_θ .

- A bank combines its customer records with external credit data.
- A hospital combines patient records with clinical or genomic data.
- An AI developer augments a proprietary training set with newly purchased data.

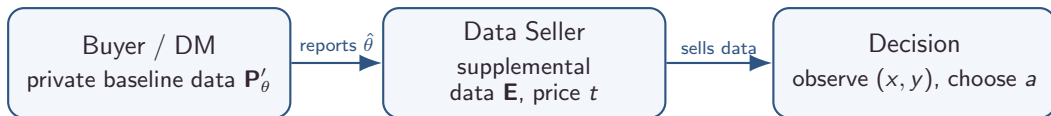
The data seller designs and prices supplementary datasets E without knowing buyers' **private datasets**.

What Makes Data Different?

Unlike standard goods, data has four distinctive features:

- **Multidimensional.** A dataset can improve several margins of a decision problem at once.
- **Rigid design.** These improvements must arise from a *single feasible information structure*; they cannot generally be designed or allocated dimension by dimension.
- **Combinatorial.** New data may be highly valuable when the buyer knows little, but nearly redundant when the buyer already has similar information.
- **Flexible, non-contractible post-purchase use.** After purchase, the buyer privately decides how to process and combine his own data with the seller's data.

**Data creates multidimensional scope for screening,
but hidden post-purchase processing limits how finely horizontal heterogeneity can be monetized.**



Buyer / Decision Maker

Decision problem: (Ω, μ, A, u) ,
where $u : A \times \Omega \rightarrow \mathbb{R}$, $\mu \in \Delta(\Omega)$.

Private type: baseline dataset

$$\mathbf{P}'_\theta : \Omega \rightarrow \Delta(\mathcal{X}), \quad \theta \in \Theta.$$

**Ex ante: no signal is realized
before contracting.**

Designer / Data Seller

Supplemental dataset:

$$\mathbf{E} : \Omega \rightarrow \Delta(\mathcal{Y}).$$

Direct mechanism:

$$\{(\mathbf{E}_\theta, t_\theta)\}_{\theta \in \Theta}.$$

Baseline and supplemental data are
conditionally independent given state.

Decision & Data Value

After purchase, the buyer observes

$$x \sim \mathbf{P}'_\theta, \quad y \sim \mathbf{E}.$$

The buyer chooses

$$a^*(x, y) \in \arg \max_{a \in A} [u(a, \omega) \mid x, y].$$

Value of data is incremental:

$$\mathbb{E}[u(a, \omega) \mid (\mathbf{E}, \mathbf{P}'_\theta)] - \mathbb{E}[u(a, \omega) \mid \mathbf{P}'_\theta].$$

[1] How to reduce the double deviation in communication equilibrium and reformulate the design.

Encoding Principle

1. Reduction: encode all double deviations into garblings of the buyer's private dataset.
2. Reformulation: a (nested) bundling problem under Blackwell Monotonicity (BM)

[2] In hypothesis testing setting: duality and optimal menu structure

BM coarsens the optimal menu to the posted price for fully informative dataset in many settings.

[3] We conceptually compare and mathematically extend Bergemann, Bonatti, and Smolin (2018)

2 Encoding Principle

The $2 \times n \times 2$ Environment

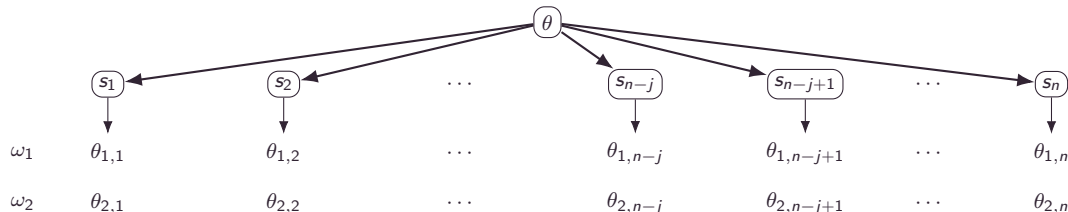
$$\Omega = \{\omega_1, \omega_2\}, \quad \Pr(\omega_i) = \frac{1}{2}, \quad A = \{a_1, a_2\}, \quad u(a_j, \omega_i) = \mathbf{1}\{i = j\}.$$

Private signals $\mathcal{X} = S_n = \{s_1, \dots, s_n\}$.

Represent the private experiment by $\theta_{ik} := \Pr(\omega_i, s_k)$.

$$\Theta_n = \left\{ \theta \geq 0 : \sum_{k=1}^n \theta_{1k} = \sum_{k=1}^n \theta_{2k} = \frac{1}{2} \right\} = \frac{1}{2} \Delta \times \frac{1}{2} \Delta.$$

Without supplementary data, $b(\theta) = \sum_{k=1}^n \max\{\theta_{1k}, \theta_{2k}\}$.



We can allow type-dependent signal space with different bounded cardinality, $|S_\theta|$.

Our result does not rely on uniform prior and matching utility.

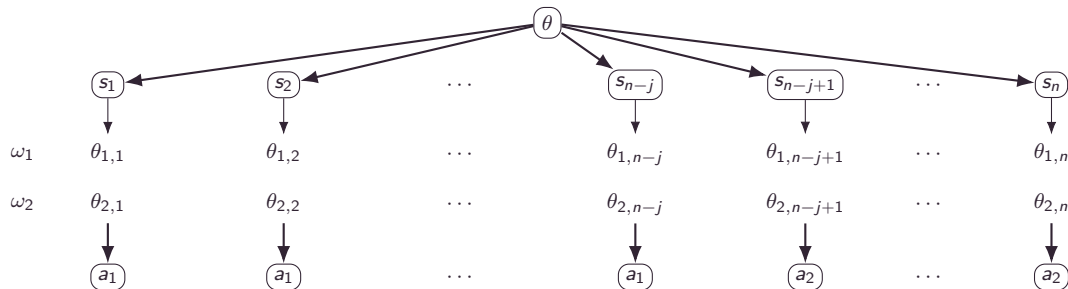
Supplementary Data

Report θ receives $(X(\theta), t(\theta))$. Order θ by likelihood ratios.

Revelation principle: realization j recommends the cutoff strategy

$$c_j^\theta(q) = \left(\underbrace{a_1, \dots, a_1}_{n-j}, \underbrace{a_2, \dots, a_2}_j \right) = \begin{cases} a_1, & q \leq n-j, \\ a_2, & q > n-j. \end{cases}$$

	c_0^θ	c_1^θ	$\dots$	c_{n-1}^θ	c_n^θ
ω_1	x_{10}	x_{11}	$\dots$	$x_{1,n-1}$	0
ω_2	0	x_{21}	$\dots$	$x_{2,n-1}$	x_{2n}



Set likelihood ratio as ∞ or 0 for ω_1 or ω_2 fully revealing signals.

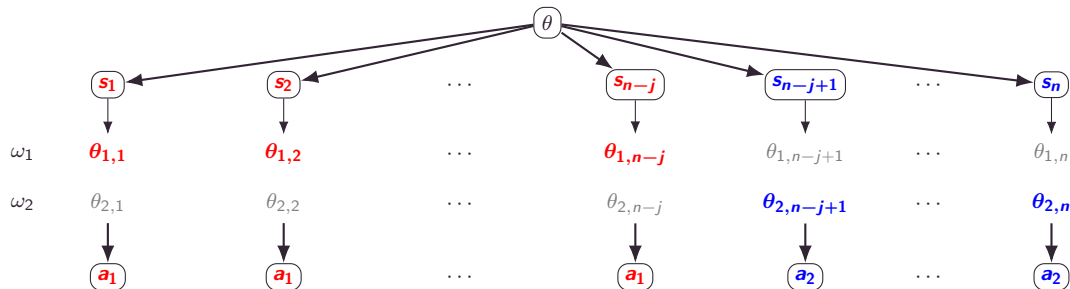
A Bundling Representation

For report r , obedient use of $X(r)$ gives type θ the gross value

$$u_{\theta}^o(r) = \sum_{q=1}^n \left[\theta_{1,q} \sum_{j=0}^{n-q} x_{1j}(r) + \theta_{2,q} \sum_{j=n-q+1}^n x_{2j}(r) \right] := \sum_{j=0}^n x_{1j}(r) v_{\theta}(A_j) + \sum_{j=0}^n x_{2j}(r) v_{\theta}(B_j).$$

where

$$v_{\theta}(A_j) = \sum_{q=1}^{n-j} \theta_{1,q}, \quad v_{\theta}(B_j) = \sum_{q=n-j+1}^n \theta_{2,q}, \quad j = 0, \dots, n.$$



Stochastic Choice over Two Nested Bundling

Let a_{ik} denote choosing action a_i after private signal s_k . The value of a_{ik} is θ_{ik} .

The value for bundling $\{a_{ik}\}_{k \in K}$ is linearly additive.

Two bundle chains.

$$\begin{aligned}\{a_{11}, a_{12}, \dots, a_{1n}\} &= A_0 \supset \{a_{11}, a_{12}, \dots, a_{1n-1}\} = A_1 \supset \dots \supset A_{n-1} = \{a_{11}\} \supset A_n = \emptyset, \\ \emptyset &= B_0 \subset B_1 = \{a_{2n}\} \subset B_2 = \{a_{2n}, a_{2n-1}\} \subset \dots \subset B_n = \{a_{21}, a_{22}, \dots, a_{2n}\}.\end{aligned}$$

We represent the agent's utility as choosing $x_i = (x_{ik}) \in \Delta$ over $v(A) = (v_\theta(A_k))$ and $v(B) = (v_\theta(B_k))$

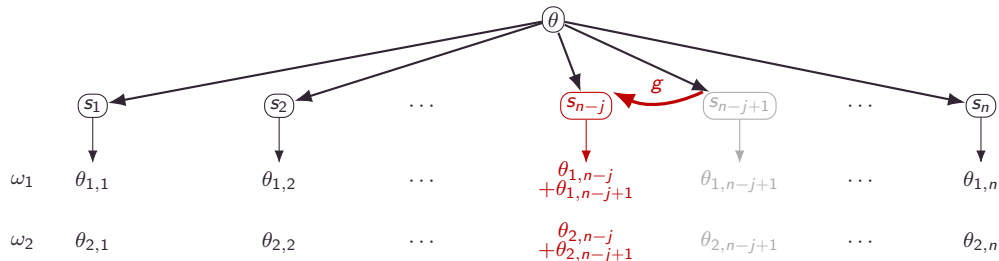
$$u^o(\theta) := x_1 \cdot v_\theta(A) + x_2 \cdot v_\theta(B) - t(\theta) - \max_k \{v_\theta(A_k) + v_\theta(B_k)\}$$

However, the buyer can still misreport and privately disobey, also with type-dependent outside option.

Garbling of Private Dataset

Let $\mathcal{G}_n := \{g : S_n \rightarrow S_n\}$. A deterministic g merges/relabels private signals and induces

$$(T_g \theta)_{i\ell} := \sum_{k: g(k)=\ell} \theta_{ik}.$$



Remark. We can relax to stochastic $g : S_n \rightarrow \Delta(S_n)$, which **garbles** the private experiment.

The merged signal should be re-ordered by likelihood ratio.

Encoding Double Deviations into Garblings

For a contract $X(r)$, we have two forms of payoffs

$u_\theta^o(r)$: obeying the report- r recommendation, and $u_\theta^*(r)$: optimally processing the purchased data.

Encoding double deviations into garblings

Double deviations can be represented as optimal garbling of data, i.e. $u_\theta^*(r) = \max_{g \in \mathcal{G}_n} u_{T_g \theta}^o(r)$.

recieve private signal θ and report $r \iff$ garble to $T_g \theta$ + obey report- r recommendation

Encoding Principle

For a contract $X(\theta)$, define the incremental net utility for θ is $u(\theta) := u_\theta^o(\theta) - t(\theta) - b(\theta)$.

Truth-telling. For every $\theta, r \in \Theta_n$, $u(\theta) \geq u_\theta^o(r) - t(r) - b(\theta)$.

Blackwell Monotonicity. For any $G : S_n \rightarrow S_n$, the overall net utility is monotone in Blackwell order

$$u(\theta) + b(\theta) \geq u(T_G \theta) + b(T_G \theta). \quad (\text{BM})$$

Incentive Compatibility. For every $\theta, r \in \Theta_n$,

$$u(\theta) \geq u_\theta^*(r) - t(r) - b(\theta). \quad (\text{IC})$$

Encoding Principle

$$\text{IC} \iff \text{Truth-telling} + \text{Blackwell Monotonicity}.$$

BM: buyer-side flexibility constrains seller-side **rent differentiation**: $u(T_G \theta) - u(\theta) \leq b(\theta) - b(T_G \theta)$.

3 Hypothesis Testing

We further specialize a two private signal case $\mathcal{X} = \{s_1, s_2\}$. s_1 : accept null ω_1 , s_2 reject null ω_2

Baseline Dataset ($p_1(\theta) + p_2(\theta) \geq 1$)

$\mathbf{P}'_{\theta}$	s_1	s_2
ω_0	$p_1(\theta)$	$1 - p_1(\theta)$
ω_1	$1 - p_2(\theta)$	$p_2(\theta)$

Type I Error $\alpha_{\theta} = \frac{1}{2}(1 - p_1(\theta))$

Type II Error $\beta_{\theta} = \frac{1}{2}(1 - p_2(\theta))$

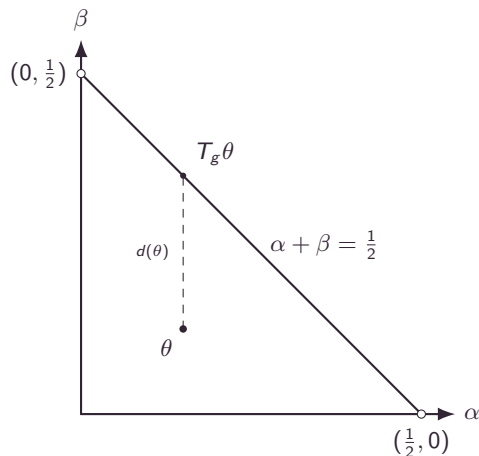
private type: $\theta = (\alpha_{\theta}, \beta_{\theta}) \in \mathbb{R}_+^2$, $\alpha_{\theta} + \beta_{\theta} \leq \frac{1}{2}$

design supplement dataset = allocate **statistical error reduction** $(x_1(\theta), x_2(\theta)) \in [0, 1]^2$ to $\theta = (\alpha_{\theta}, \beta_{\theta})$

incremental net value $u(x) = \alpha x_1 + \beta x_2 - t(\theta) = \theta \cdot x - t(\theta)$.

The model is isomorphic to selling information to agents with private decision problem, i.e. $u_i(a, \omega)$ -vector is the private information.

Triangular Type Space Θ



$$\Theta = \{(\alpha, \beta) \mid \alpha + \beta \leq \tfrac{1}{2}\} \quad H = \{(\alpha, \beta) \mid \alpha + \beta = \tfrac{1}{2}\}$$

For any $\theta \in \Theta$ and merging g , $T_g \theta \in H$.

$$d(\theta) = \tfrac{1}{2} - \alpha - \beta = \|\theta - T_g \theta\|_1 \text{ for any merging } g.$$

$\alpha + \beta$: overall statistical error, (α, β) : horizontal preference.

Designer's Problem: from Allocation to Utility Design

The seller solves

$$\max_{x,t} \int_{\Theta} t(\theta) dF(\theta)$$

subject to

$$\text{IC: } u(\theta) \geq \theta \cdot x(\theta') - t(\theta'), \quad \forall \theta, \theta' \in \Theta,$$

$$\text{IR: } u(\theta) \geq 0, \quad \forall \theta \in \Theta,$$

$$\text{BM: } u|_H \equiv r \text{ and } u(\theta) \geq r - d(\theta) \text{ for some } r \in [0, 1/2], \quad \forall \theta \in \Theta.$$

By Rochet (1987), the mechanism can be represented by u and designer's objective is

$$\max_{u \in \mathcal{U}_0} \int_{\Theta} u d\nu$$

where

$$\mathcal{U}_0 = \{u \mid (\text{IC}), (\text{IR}), (\text{BM}) \text{ holds and } x(\theta) \in \partial u(\theta), t(\theta) = \theta \cdot x(\theta) - u(\theta)\},$$

$\nu := \delta_0 - (3f(\theta) + \theta \cdot \nabla f(\theta))d\theta$ is the multi-dimensional version of virtual surplus

Implications of Blackwell Monotonicity

At an optimum we normalize $u(0,0) = 0$. The feasible utility set $\mathcal{U}_0$ is

u convex,	(Convexity)
u coordinatewise nondecreasing,	(Monotonicity)
$u _H = r$ and $u(\theta) \geq r - d(\theta) := r + \alpha + \beta - \frac{1}{2}, \quad \forall \theta \in \Theta.$	(BM)
$u(\theta) - u(\theta') \leq \ \theta - \theta'\ _1 \quad \forall \theta, \theta' \in \Theta.$	(ℓ_1 -Lipschitz)

(ℓ_1 -Lipschitz) is implied by (BM) due to the **triangular type space** and **no-distortion-at-the-frontier**.

Implications of Blackwell Monotonicity

1. Horizontal differentiation is disciplined by vertical informativeness.
2. This reduces the duality problem from a transport/coupling problem (Dasklakis et.al 2017) to a stochastic order optimization/persuasion problem (Yang and Yang, 2026).

Primal and Dual: Capped Allocation and Generalized Persuasion

Recall feasible utility function space $\mathcal{U}_0 := \{u : u \text{ convex, nondecreasing, } u|_H = r\}$.

Primal: Capped Allocation

$$V = \max_{\substack{u \in \mathcal{U}_0 \\ u(\theta) \geq r - d(\theta) \\ r \in [0, 1/2]}} \int_{\Theta} u \, d\nu.$$

Dual: Generalized Persuasion

$$V = \min_{\lambda \in \mathcal{M}_+(\Theta)} \left\{ \int_{\Theta} d(\theta) \, d\lambda(\theta) : \lambda \succeq_{\mathcal{U}_0} \nu \right\},$$

where

$$\lambda \succeq_{\mathcal{U}_0} \nu \text{ iff } \int_{\Theta} u \, d(\lambda - \nu) \geq 0, \quad \forall u \in \mathcal{U}_0$$

Both optima are attained;

Strong duality follows from Gretskey–Ostroy–Zame (2002) and “perturbed” Kleiner–Manelli (2019).

Geometric Operations behind the Stochastic Order

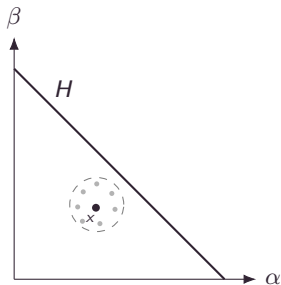
Recall the stochastic order is defined by test function class

$$\mathcal{U}_0 := \{u : u \text{ convex, } + \text{ nondecreasing, } + u|_H = r\}$$

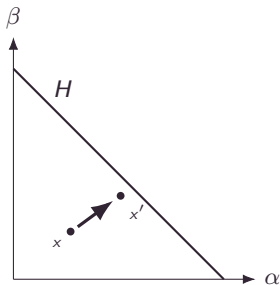
Extending Strassen (1965), the induced stochastic order admits three geometric operations:

MPS + **upward movement** + **frontier reshuffling**

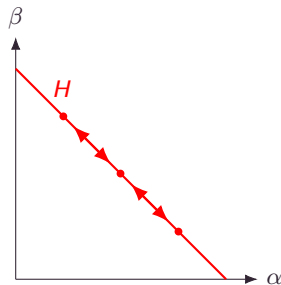
Mean-preserving spread



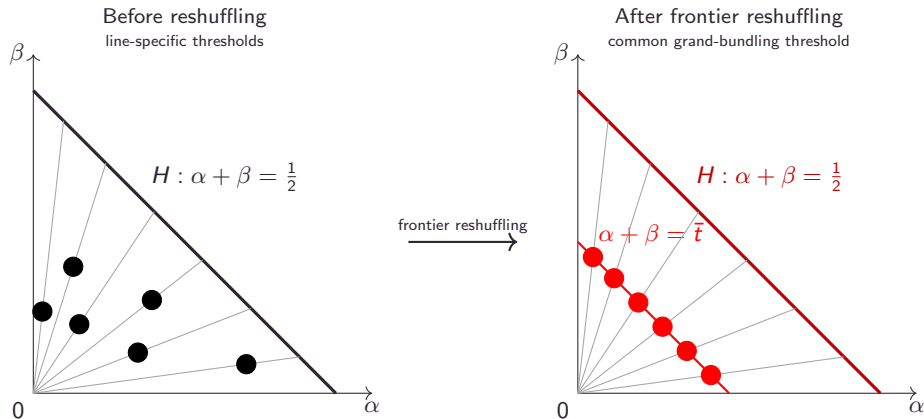
Upward movement



Frontier reshuffling



Blackwell Monotonicity coarsens the menu structure



Frontier reshuffling aligns ray-specific posted price thresholds into a common cutoff

→ coarsening horizontal differentiation into grand bundling with price $\bar{t}$

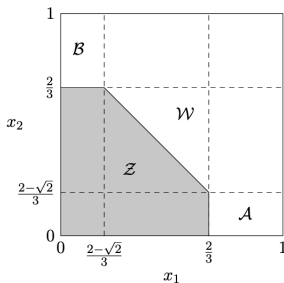
Increasing virtual value implies no-haggling?

A long discussion: McAfee and McMillan (1988), Thanassoulis (2004), Pavlov (2010), ...

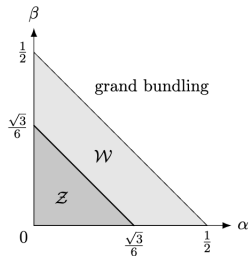
grand bundling $\mathcal{W} = (1, 1)$, exclusion $\mathcal{Z} = (0, 0)$, horizontal differentiation $\mathcal{A} = (1, 0), \mathcal{B} = (0, 1) \rightarrow \phi$

Optimality of grand bundling I: no haggling under increasing virtual value

Grand bundling is optimal if $3f(\theta) + \theta \cdot \nabla f(\theta) \geq 0$ on Θ .



(a) Optimal mechanism on $[0, 1]^2$ with i.i.d $U[0, 1]$.



(b) Grand bundling above $\alpha + \beta = \sqrt{3}/6$.

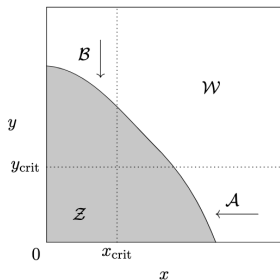
Infinitely many options in the optimal menu?

A long discussion: Hart and Reny (2015), Daslakis et.al (2017), Niemeyer and Lahr (2026), ...

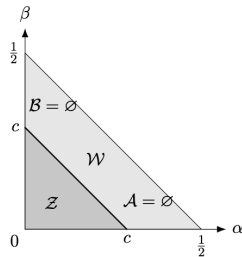
$\mathcal{A} : (1, s_2(\beta))$, $\mathcal{B} : (s_1(\alpha), 1)$ horizontal differentiation with **infintely many options** $\rightarrow \phi$

Optimality of Grand Bundling II: Z-menu collapses to posted price

Grand bundling is optimal under DDT(2017)'s sufficient conditions for the Z-mechanism.



(a) DDT(2017): canonical partition



(b) Our result: grand bundling.

Add constants to the frontier-gap cone:

$$\widehat{\mathcal{K}} := \{\phi : \phi \text{ concave, coordinatewise nonincreasing, and constant on } H\}.$$

For $0 \leq \psi \leq d$, define its $\widehat{\mathcal{K}}$ -envelope

$$\mathcal{E}\psi(x) := \inf\{\phi(x) : \phi \in \widehat{\mathcal{K}}, \phi \geq \psi\} = \max_{\substack{m \in \Theta \\ m \geq x}} \text{cav}_{\Theta} \psi(m).$$

Pointwise Envelope (Yang², 2026)

The seller's problem becomes

$$V = \max_{0 \leq \psi \leq d} \left\{ \int_{\Theta} \psi d\nu^{-} - \int_{\Theta} \mathcal{E}\psi d\nu^{+} \right\}.$$

Next questions: pooling space? extreme points? (Kleiner, Moldvanu, Strack, Whitmeyer, 2026)

4 Discussion: Bergemann, Bonatti, and Smolin (18 AER)

Bergemann, Bonatti, Smolin (2018 AER) “The Design and Price of Information”

designer prices and designs information structure E , to screen agents with heterogenous belief μ

	BBS (2018)	HZZ (2026)
Timing	Interim: one signal realization before contracting	Ex ante: only know the structure of private dataset before contracting
Private type	Interim belief $\mu_\theta \in \Delta(\Omega)$	posterior distribution $\tau_\theta \in \Delta(\Delta(\Omega))$
Error	One-sided error: $(\alpha, 0)$ or $(0, \beta)$	Two-dimensional error: (α, β)
Application	reduce existing decision mistakes	optimize prediction models

Simplification of BBS (2018)

Type $\Theta = \{(\alpha, 0)\} \cup \{(0, \beta)\}$ **Allocation** $x_1(\alpha), x_2(\beta)$

Optimization

$$\max \int t_\alpha F_1(d\alpha) + \int t_\beta F_2(d\beta)$$

subject to

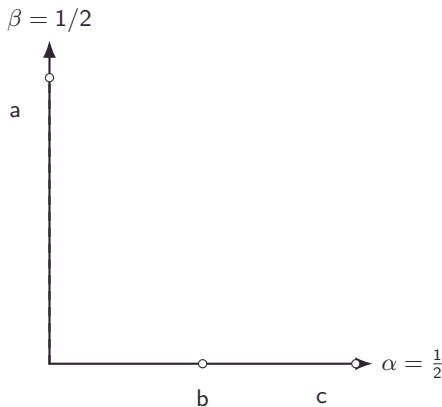
■ x_1, x_2 monotone & envelope formula

■

$$\int_0^1 x_1 d\alpha = \int_0^1 x_2 d\beta$$

Solution

$$x_1 = \lambda \mathbf{1}_{\alpha \geq b} + (1 - \lambda) \mathbf{1}_{\alpha \geq c} \quad x_2 = \mathbf{1}_{\beta \geq a}$$



Optimal menu of information selling (BBS 2018)

The optimal menu has at most three menu options: fully informative, null, partially informative.

Data is multidimensional, but private data processing can make screening effectively one-dimensional.

1 Selling data is screening with hidden information processing.

Buyer's private dataset determines both willingness to pay and how purchased information is used.

2 Encoding Principle.

Data selling=Bundling & Hidden post-purchase processing can be encoded as Blackwell garbling:

$$\text{IC} \iff \text{Truth-telling} + \text{Blackwell Monotonicity}.$$

3 BM makes horizontal heterogeneity harder to monetize, coarsening the optimal menu.

It links horizontal error composition to vertical informativeness, and can collapse multidimensional screening to a posted price for fully informative data.